

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5 TOTAL MARKS: 50 Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I P.Paneendra (192321072) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Paneendra

Annexure A
PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.
2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.
3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction. 4.
Design pseudocode for an Intrusion Detection System (IDS) that uses both signature-based and anomaly-based detection to identify suspicious network activity.
5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

PSEUDOCODE

1. Pseudocode for Simplified TLS Handshake

Objective: Simulate client request, server authentication using a digital certificate, session-key generation, and secure data exchange.

START

Client → Server: Send ClientHello

- Send supported TLS versions

- Send supported cipher suites

- Send random value

Server → Client: Send ServerHello

- Select TLS version

- Select cipher suite

- Send server random value

Server → Client: Send Digital Certificate

Client:

- Verify server certificate

- Check certificate validity

- Check trusted Certificate Authority

IF certificate is valid THEN

 Continue handshake

ELSE

 Display "Server authentication failed"

 Terminate connection

END IF

Client and Server:

 Perform key exchange

 Generate shared session key

Client → Server:

 Send encrypted handshake confirmation

Server → Client:

 Send encrypted handshake confirmation

IF both confirmations are valid THEN

 Establish secure TLS session

ELSE

 Terminate connection

END IF

Client:

 Encrypt application data using session key

Send encrypted data to Server

Server:

Decrypt received data

Process data

Encrypt response using session key

Send encrypted response

Client:

Decrypt response

END

Result: A secure session is established after authentication and session-key generation, and application data is exchanged in encrypted form.

2. Pseudocode for Firewall Rule Engine

Objective: Filter incoming packets according to IP address, port, and protocol.

START

Define Firewall Rules

Rule 1:

IP = 192.168.1.10
Port = 80
Protocol = TCP
Action = ALLOW

Rule 2:

IP = 10.0.0.5
Port = 23
Protocol = TCP
Action = BLOCK

FOR each incoming packet

Read Source_IP
Read Destination_IP
Read Destination_Port
Read Protocol

Rule_Matched = FALSE

FOR each firewall rule

IF Source_IP matches rule IP AND
Destination_Port matches rule Port AND
Protocol matches rule Protocol THEN

Rule_Matched = TRUE

IF rule Action = ALLOW THEN

 Allow packet

 Log "Packet Allowed"

ELSE IF rule Action = BLOCK THEN

 Block packet

 Log "Packet Blocked"

END IF

 BREAK

END IF

END FOR

IF Rule_Matched = FALSE THEN

 Apply default policy

 IF Default_Policy = ALLOW THEN

 Allow packet

 ELSE

 Block packet

 END IF

END IF

END FOR

STOP

Result: The firewall compares every incoming packet with predefined rules and either allows or blocks the traffic.

3. Pseudocode for Malware Behavior Classification

Objective: Classify malware as a **virus, worm, or Trojan** based on propagation, file modification, and user interaction.

START

Receive suspicious program/activity

Check propagation method

Check file modification behavior

Check user interaction

IF malware spreads automatically

 AND scans or connects to other systems

 AND does not require user action THEN

 Classification = "WORM"

ELSE IF malware attaches itself to files
OR modifies existing executable files
AND spreads when infected files are executed THEN

Classification = "VIRUS"

ELSE IF malware appears as legitimate software
AND requires user execution or interaction
AND performs malicious activities THEN

Classification = "TROJAN"

ELSE

Classification = "UNKNOWN"

END IF

Display Classification

IF Classification = "WORM" THEN
Isolate affected network/system
Block suspicious network connections

ELSE IF Classification = "VIRUS" THEN

Scan infected files
Quarantine infected files

ELSE IF Classification = "TROJAN" THEN

Terminate suspicious process
Quarantine malicious program

END IF

Log malware activity

STOP

Result: The malware is classified according to how it spreads, modifies files, and interacts with the user.

4. Pseudocode for Signature-Based and Anomaly-Based IDS

Objective: Detect suspicious network activity using both known attack signatures and abnormal behavior.

START

Load attack signatures
Create normal network behavior profile

WHILE network is active

Capture incoming network packet

Extract:

Source IP

Destination IP

Port

Protocol

Packet contents

Traffic behavior

Signature_Match = Compare packet
with known attack signatures

Anomaly_Score = Compare packet behavior
with normal behavior profile

IF Signature_Match = TRUE THEN

Alert = "Known Attack Detected"

Severity = HIGH

Log attack details

ELSE IF Anomaly_Score > Threshold THEN

Alert = "Anomalous Activity Detected"

Severity = MEDIUM/HIGH

Log suspicious activity

ELSE

Alert = "Normal Traffic"

END IF

END WHILE

STOP

Detection process

Network Traffic



Packet Capture



Signature Detection



Anomaly Detection



Risk Evaluation



Normal / Suspicious / Attack



Generate Alert and Log

Result: Signature-based detection identifies known attacks, while anomaly-based detection identifies unusual behavior that may represent unknown attacks.

5. Pseudocode for Intrusion Prevention System (IPS)

Objective: Monitor network traffic continuously, detect malicious requests, automatically block them, and record intrusion attempts.

START

Load attack signatures

Load security rules

Set anomaly threshold

WHILE network is active

 Capture incoming request

 Extract:

 Source IP

 Destination IP

Port
Protocol
Request data

Check request against attack signatures

Calculate risk score

IF known malicious signature is detected THEN

Block request

Log:

Source IP

Destination IP

Attack type

Time

Action = BLOCK

Generate security alert

ELSE IF risk score > threshold THEN

Block request

Log suspicious activity

Generate security alert

ELSE

Allow request

Log normal traffic if required

END IF

END WHILE

STOP

IPS process

Network Traffic



Traffic Monitoring



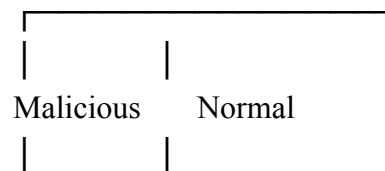
Packet Inspection

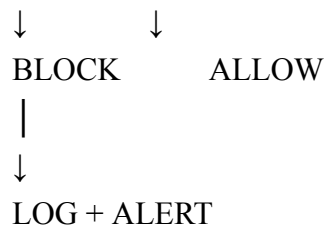


Signature + Anomaly Detection



Threat Decision





Result: Unlike an IDS, which mainly detects and reports attacks, an IPS actively **blocks malicious traffic in real time** while maintaining logs for security analysis.